Intraday Pairs Trading with Machine Learning Filters:

An Empirical Study of the Post-Decimalization Decline on the S&P 500

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Abstract

We benchmark nine variants of an intraday statistical arbitrage pairs trading strategy on a sample of 39 S&P 500 large caps over 2020–2026, faithfully reproducing the methodology of Stübinger and Bredthauer (2017) and extending it with two machine-learning pair-selection filters: K-means clustering on formation-window features and a rolling autoencoder k-nearest-neighbor filter à la Jung et al. (2024). Three central findings emerge. First, all nine method variants exhibit a positive gross Sharpe (best: *ae+correlation* at 1.95) but a *negative* net Sharpe after realistic transaction costs (best: *kmeans+cointegration* at -1.48). Second, a per-round-trip decomposition shows that the statistical edge per trade has compressed to 0.65–3.69 basis points, two orders of magnitude below the ~ 100 bps implied by Stübinger's 1998–2015 sample, while the cost layer we apply (~ 10 bps/round-trip) is *more conservative* than the implicit cost structure of the literature. Third, Fama–French five-factor regressions confirm statistically significant *negative* daily alphas (-7.4 to -9.9 bps, $t \in [-17, -4]$) across the entire method grid — in direct contrast with Stübinger's reported $+16$ bps daily alpha ($t = 13.77$). These results are *consistent with* — and extend beyond 2015 — the declining-profitability literature of Do and Faff (2012) and Landgraf (2016). Machine-learning filters partially mitigate the decline (the K-means cointegration filter lifts gross per round-trip by a factor of $5.7\times$ over baseline cointegration) but are not sufficient to restore profitability on our sample.

Keywords: Statistical arbitrage; pairs trading; high-frequency; machine learning; K-means clustering; autoencoder; transaction costs; Fama–French alpha.

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1 Introduction

Pairs trading — the simultaneous opening of a long and short position in two historically related assets — is one of the canonical market-neutral strategies of the modern equity hedge fund industry. Since Gatev et al. (2006)'s landmark study documenting an 11% annual excess return over 1962–2002, the strategy has become an archetype of statistical arbitrage, taught in every quantitative finance curriculum. Its intellectual appeal is rooted in a clear economic narrative: cointegrated or economically related securities exhibit mean-reverting relative-price dynamics that can be exploited without directional market exposure.

Yet two decades of subsequent research have documented a persistent *decline* in pairs trading profitability. Do and Faff (2010, 2012) show that the simple Gatev rules, once applied net of realistic transaction costs, lose statistical significance after the early 2000s. Bowen and Hutchinson (2016) reach similar conclusions on the FTSE. Most strikingly, Landgraf (2016) documents that between 2008 and 2016, virtually all variants of pairs trading — distance, cointegration, copulas — *fail to generate positive returns* once 50 bps per round-trip transaction costs are applied, a finding consistent with the broad literature on arbitrage decay under high-frequency competition (Khandani and Lo, 2007; Krauss, 2017).

Against this backdrop, Stübinger and Bredthauer (2017) propose one of the most optimistic modern defenses of high-frequency pairs trading: working on S&P 500 constituents over 1998–2015 with minute-level data and varying Bollinger bands, they report up to 50.5% annualized net returns and a Fama–French five-factor alpha of +40.3% per year ($t = 13.77$). Their strategy "EV" — Euclidean distance pair selection with varying volatility thresholds — is their poster child. The question our study addresses is natural:

> *Does the Stübinger methodology survive when applied to the 2020–2026 sample, and can modern machine-learning pair selection restore the profitability?*

Our contribution is threefold. First, we reproduce the Stübinger framework on a 39-ticker S&P 500 sub-sample, building out the full pipeline from raw M1 data filtering to Fama–French alpha estimation, with explicit per-round-trip cost accounting. Second, we extend the framework with two ML filters — rolling K-means clustering on formation features (Sarmento and Horta, 2020) and a rolling autoencoder k-NN latent-space filter (Jung et al., 2024). Third, we provide a granular diagnostic (gross vs. net per round-trip, k-threshold sensitivity, sub-period stability, Fama–French alpha) that isolates the mechanism behind our results.

The main finding is that the decline documented by Landgraf (2016) *extends beyond 2015* into the post-COVID period, and is structural: the gross statistical edge per trade has compressed from approximately 100 bps in Stübinger's 1998–2015 sample to 1–3 bps on our 2020–2026 sample. Machine-learning filters systematically improve the edge (the K-means cointegration filter delivers a 5.7× improvement over baseline) but remain insufficient to restore net profitability once transaction costs are applied.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents the methodology, including the ML filter design

and the cost model. Section 5 reports results with extensive diagnostics. Section 6 discusses the findings and their implications. Section 7 outlines directions for future research.

2 Literature review

2.1 Classical pairs trading: the Gatev framework

Gatev et al. (2006) formalize the "distance" approach, where pairs are selected during a formation period by minimizing the sum of squared normalized price distances:

$$(a^*, b^*) = \arg\min_{(a,b)} \sum_{t \in \text{formation}} (\tilde{P}_{a,t} - \tilde{P}_{b,t})^2, \tag{1}$$

with $\tilde{P}$ denoting the cumulative return-normalized price. Trading positions are opened when the spread deviates by two standard deviations from its formation-period mean and closed when it re-crosses the mean. Over 1962–2002, this simple rule delivers 11% annualized excess return with minimal market exposure.

Vidyamurthy (2004) introduces the cointegration-based variant, in which pairs are selected via the Engle–Granger two-step procedure: a candidate pair is accepted if the OLS residual $\varepsilon_t = P_{a,t} - (\alpha + \beta P_{b,t})$ has a stationary ADF test. The hedge ratio β becomes the optimal statistical weight on leg B, which for a cointegrated pair guarantees a mean-reverting spread. This framework is theoretically appealing but empirically more sensitive to non-stationarity and structural breaks (Krauss, 2017).

2.2 High-frequency extensions and the profitability decline

Bowen et al. (2010) apply the Gatev framework to 1-minute FTSE data and show that transaction costs dominate the gross signal for small-cap pairs. Do and Faff (2010, 2012) rigorously evaluate the robustness of the Gatev strategy under increasingly realistic cost assumptions (10 bps to 50 bps per round-trip) and find that by the 2000s, most of the naive distance strategy's profitability has been eroded.

Landgraf (2016) conducts the most comprehensive study of this decline on NASDAQ using 2000–2016 data, applying five pair-selection methodologies (distance, cointegration, empirical copula, parametric copula, time-varying copula). Figure 9 of that thesis, reproduced in spirit below , shows a clear post-2008 collapse: the annualized Sharpe ratio of every variant drops from 5–9 before the crisis to near zero or negative thereafter, under a 50 bps per round-trip cost scheme. Landgraf concludes: "*from 2008 to 2016, all strategies become unprofitable with negative average returns*".

2.3 The Stübinger defense

Against this skeptical backdrop, Stübinger and Bredthauer (2017) offer a renewed positive case. Using the full S&P 500 over 1998–2015 at 1-minute frequency, with three alternative threshold policies (static, varying, reverting) combined with four selection rules (Euclidean distance, cointegration, fluctuation, correlation), they obtain strategy EV (distance + varying thresholds) with +50.5% annualized return after costs, Sharpe ratio 8.14, and a Fama–French five-factor alpha of +16 bps per day ($t = 13.77$). Their Table 8 reports consistent annual Sharpe ratios above 4 throughout 1998–2015 for strategy EV.

Two methodological choices are key to the Stübinger setup. First, transaction costs are modeled as 5 bps per share per half-turn, all-in (spread + commission + impact). Second, they use "varying" Bollinger thresholds that adapt to rolling spread volatility, which permits entry on relative divergences even during low-volatility regimes. Their implicit round-trip cost is therefore ~ 20 bps per pair.

2.4 Machine learning in pair selection

Sarmento and Horta (2020) propose K-means clustering on formation-window features (volatility, skewness, autocorrelation) as a pre-filter that restricts pair candidates to economically similar stocks. They report statistical improvements on cointegration-based strategies. Jung et al. (2024) introduce an autoencoder-based approach ("Approach B"): formation-window prices are embedded into a low-dimensional latent space, and pairs are selected from the k-nearest-neighbor set in latent space. Both approaches address a common weakness of classical pair selection: the reliance on linear features that may miss non-linear economic relationships.

Our study unifies these approaches under a common backtesting framework and exposes their marginal contribution over naive baselines under conservative cost assumptions.

3 Data

3.1 Source and coverage

We work with 1-minute mid-prices from ICMarkets (MetaTrader) for 39 S&P 500 large caps spanning 11 GICS sectors. The raw files cover January 16, 2020 to March 30, 2026, a total of 1 558 trading days. Table 1 reports the universe; Table 2 summarizes data quality after the filtering pipeline described below.

3.2 Cleaning and alignment

We apply a standard high-frequency data cleaning protocol:

1. **Day-level flagging:** For each ticker-day, we flag days with fewer than 300 bars (from a nominal ~ 390 minutes per session) or with intraday return jumps exceeding a Tukey fence

Table 1: Ticker universe (39 S&P 500 large caps, by sector).

Sector	Tickers
Information Technology	AAPL, MSFT, NVDA, ORCL, CSCO, IBM, AVGO, TXN, QCOM, AMD, INTC
Communication Services	GOOG, NFLX
Consumer Discretionary	AMZN, TSLA, NKE, SBUX
Consumer Staples	PG, KO, PEP, MDLZ
Energy	XOM, CVX, COP, SLB
Financials	JPM, BAC, WFC, C, MS, USB, V
Health Care	JNJ, UNH, MRK, GILD, CVS
Industrials	BA, GE

($|r_t| > Q_3 + 3 \cdot \text{IQR}$).

2. **Completeness filter:** We retain only tickers with $\geq 85\%$ clean-day rate. This eliminates tickers with sporadic data issues.

3. **Alignment:** All ticker series are aligned on a common M1 intraday grid; missing bars are forward-filled within the same day only.

4. **Mid-price reconstruction:** Daily closes are derived from the last clean bar of each day, for the sector-feature analysis in Step 3.

After filtering, median completeness is 92.11% (min 85.04%, max 96.21%). The universe retained is larger than any single pair selection can effectively exploit: with `TOP_N` = 10 pairs per selection method per window, we search over $\binom{39}{2} = 741$ candidate pairs each formation period.

Table 2: Data quality summary after cleaning.

Metric	**Value**
Tickers retained	39
Backtest period	2020-01-16 to 2026-03-30
Total trading days	1 558
Clean days min/max per ticker	1 325 / 1 499
Median completeness	92.11%

3.3 Comparison to Stübinger's universe

Our sample differs from Stübinger and Bredthauer (2017) in three important ways. First, universe size: we operate on 39 tickers vs. their full S&P 500 of $\sim$ 500 constituents — a factor of $\sim 12\times$ reduction in pair search space. Second, time period: our 2020–2026 sample is *five years after* the end of theirs (2015), and encompasses the COVID-19 market-disruption period (2020–2021) as well as the subsequent high-interest-rate regime. Third, we use non-overlapping walk-forward windows (`STEP` = 5 days) while Stübinger overlaps by 1 day (`STEP` = 1), yielding $\sim 5\times$ fewer formation windows. These differences are expected to reduce total capital deployed and diversification; we comment on their impact in Section 6.

4 Methodology

4.1 Walk-forward protocol

Following Gatev et al. (2006) and Stübinger and Bredthauer (2017) §4.2, each study period consists of a 10-day formation window (for pair selection and hedge-ratio estimation) followed by a 5-day trading window. The two windows are non-overlapping, yielding ~ 258 study periods over the 1 558-day sample. A 50-window warm-up is applied before any trading, to ensure that the ML filters (K-means, autoencoder) have stabilized on their rolling fit.

4.2 Pair selection methods

Distance. Following Gatev et al. (2006), for each formation window we compute the sum of squared normalized-price distances across all pairs and retain the `TOP_N` $= 10$ lowest-distance pairs. The hedge-ratio for this method is $\beta = 1$ (naive dollar-neutral).

Cointegration. Following Vidyamurthy (2004), we run an OLS regression $P_{a,t} = \alpha + \beta P_{b,t} + \varepsilon_t$ on the formation window (using daily closes, not M1, for ADF test power), test the residuals for stationarity via ADF at $p < 0.05$, and retain the top 10 pairs ranked by ADF p-value. The hedge ratio β enters both the spread definition ($S_t = P_{a,t} - \beta P_{b,t}$) and the PnL weighting (see below).

Correlation. We rank pairs by formation-window Pearson correlation, requiring $\rho_{a,b} \geq 0.7$, and retain the top 10. Hedge ratio $\beta = 1$.

4.3 Trading rule

We use *varying* Bollinger bands, following Stübinger and Bredthauer (2017) §4.3, with a rolling window of $L = 380$ bars (equivalent to approximately one trading day) to compute rolling mean μ_t and standard deviation σ_t of the spread. A position is opened when $z_t = (S_t - \mu_t)/\sigma_t$ crosses $\pm k$ and closed when z_t re-crosses zero:

$$\text{pos}_t = \begin{cases} +1 & \text{if } z_t < -k \text{ and } \text{pos}_{t-1} = 0, \\ -1 & \text{if } z_t > +k \text{ and } \text{pos}_{t-1} = 0, \\ 0 & \text{if } \text{pos}_{t-1} \neq 0 \text{ and } z_t \text{ crosses zero}, \\ \text{pos}_{t-1} & \text{otherwise.} \end{cases} \tag{2}$$

Our baseline uses $k = 2.5$, with $k \in \{2.5, 3.0, 3.5\}$ tested in Section 5.4.

4.4 Return computation

For a given pair (a, b) with position series $\{\text{pos}_t\}$ over a trading window, the bar-level pair return follows Gatev et al. (2006) eq. (2)–(3) with compounded leg weights:

$$r_t^{\text{pair}} = \text{pos}_{t-1} \cdot (w_{a,t} \cdot r_{a,t} - w_{b,t} \cdot r_{b,t}), \tag{3}$$

$$w_{a,t} = w_{a,t-1} \cdot (1 + r_{a,t-1}), \tag{4}$$

$$w_{b,t} = w_{b,t-1} \cdot (1 + r_{b,t-1}). \tag{5}$$

Weights are reset to their initial values $(w_{a,0}, w_{b,0})$ at each position open. The *initial* weights depend on the selection method:

- **Distance, correlation**: $w_{a,0} = w_{b,0} = 0.5$ (dollar-neutral \$1/\$1).
- **Cointegration**: $w_{a,0} = 1/(1 + |\beta|)$, $w_{b,0} = |\beta|/(1 + |\beta|)$ (β-weighted allocation, leg-total = \$1).

This normalization ensures that each active pair deploys exactly \$1 of gross capital regardless of method, making returns directly comparable across methods. This follows the logic of Vidyamurthy (2004)'s statistically optimal hedge while preserving Gatev's capital-neutrality principle.

Portfolio aggregation. For each bar, the portfolio return is the mean across `TOP_N` pairs, whether active or flat:

$$r_t^{\text{port}} = \frac{1}{\text{TOP_N}} \sum_{i=1}^{\text{TOP_N}} r_t^{\text{pair},i}. \tag{6}$$

This corresponds to Stübinger and Bredthauer (2017)'s "return on committed capital" (Table 7 bottom). A pair that is flat throughout a window contributes zero, effectively diluting the portfolio return. The alternative "return on employed capital" (dividing only by the number of *active* pairs per bar) gives higher figures, but is less conservative from an investor's ex-ante allocation perspective. Stübinger reports both and notes they are very close when pair activity is high (EV: 50.50% employed vs. 50.21% committed); we retain the committed version throughout.

4.5 Transaction costs

We follow Stübinger and Bredthauer (2017) §4.5 exactly: **5 basis points per share per half-turn, all-in**. This includes bid-ask spread, commission, and market impact in a single flat rate, consistent with institutional execution on US large-cap equities (Prager et al., 2012; Voya Investment Management, 2016). For pedagogical transparency, we expose the decomposition:

$$\underbrace{5\,\text{bps}}_{\text{all-in}} = \underbrace{2.5\,\text{bps}}_{\text{half-spread}} + \underbrace{1.5\,\text{bps}}_{\text{commission}} + \underbrace{1.0\,\text{bps}}_{\text{slippage}}, \tag{7}$$

but constrain the decomposition to sum exactly to the all-in rate. Applied to a pair event (opening or closing), the cost is:

$$\text{cost}_{\text{half-turn}} = (w_a + w_b) \cdot 5\,\text{bps} = 5\,\text{bps}, \tag{8}$$

since $w_a + w_b = 1$ by construction. A round-trip (open+close) therefore costs approximately 10 bps per pair, regardless of method. This is *half* of Stübinger's own effective round-trip cost (~ 20 bps) and *one-fifth* of Landgraf's conservative scheme (50 bps).

4.6 Machine-learning filters

4.6.1 K-means rolling clustering (Step 3)

Following Sarmento and Horta (2020), we construct a rolling formation-window feature vector per ticker comprising: mean return, return volatility, skewness, kurtosis, first-order autocorrelation, maximum drawdown, plus one-hot sector encoding (11 GICS sectors). We standardize features across tickers and fit a K-means clustering with K chosen by silhouette maximization. On our sample, the silhouette score peaks sharply at $K = 7$ (silhouette $= 0.563$). Clustering is re-fitted every 10 windows to adapt to regime shifts. Pairs are then restricted to intra-cluster: the three methods *kmeans+distance*, *kmeans+cointegration*, *kmeans+correlation* re-apply the same selection rule but only on same-cluster candidates.

4.6.2 Autoencoder rolling filter (Step 4)

Following Jung et al. (2024) "Approach B", we fit a rolling autoencoder on formation-window return time series. The encoder compresses each ticker's formation-window return sequence (length 3 800 bars = 10 trading days × 380 bars/day) to a 5-dimensional latent vector. The autoencoder is implemented in pure NumPy for determinism and reproducibility: a symmetric 2-layer encoder-decoder with tanh activations, trained by gradient descent with L_2 regularization ($\lambda = 10^{-4}$) for 100 epochs. We set the latent dimension to 5 based on a reconstruction-error knee analysis.

Pair candidates are formed from the k-nearest-neighbor graph in latent space ($k = 5$). The three methods *ae+distance*, *ae+cointegration*, *ae+correlation* then re-apply the distance / cointegration / correlation selection rules on this restricted candidate set. The autoencoder is re-fitted every 10 windows.

4.6.3 Stability of ML filters

A critical validation for any rolling ML filter is stability: do two consecutive refits produce similar groupings? We measure this with the Adjusted Rand Index (ARI) for K-means and the mean Jaccard similarity for AE k-NN sets:

- **K-means ARI**: 0.939 (mean across 28 consecutive refits). The clustering is essentially

stable, reflecting persistent sectoral structure.

- **AE k-NN Jaccard**: 0.508. The neighborhoods evolve moderately, capturing regime-sensitive relationships, but retain a meaningful fraction of the previous refit's neighbors.

These figures are broadly consistent with the stability literature on rolling unsupervised methods (Jung et al., 2024).

4.7 Performance metrics

We report all standard performance metrics on the daily return series (aggregated from M1 bars by sum over the day), with annualization factor $\sqrt{252}$:

- **Sharpe ratio**: Sharpe $= \frac{\mu_d}{\sigma_d}\sqrt{252}$.
- **Sortino ratio**: Sortino $= \frac{\mu_d}{\sigma_d^-}\sqrt{252}$, where σ_d^- is the downside volatility.
- **Calmar ratio**: Calmar $= \frac{\mu_d \cdot 252}{|\text{MaxDD}|}$.
- **Win rate**: fraction of non-zero daily returns that are positive.
- **Cost drag**: cumulative gross_pnl − net_pnl.

We also report a per-round-trip decomposition (gross, cost, net per RT, in basis points) that proves essential for diagnosis (see Section 5.2.1).

5 Results

We organize the results in four subsections. Section 5.1 presents the baseline "no-cost" results that isolate the gross statistical edge. Section 5.2 documents the impact of transaction costs, and the per-round-trip diagnostic that pinpoints the root cause. Section 5.3 evaluates the ML filters against the baseline. Section 5.4 examines the k-threshold sensitivity. Section 5.5 documents sub-period stability. Section 5.6 reports Fama–French alphas.

5.1 Baseline: the gross statistical edge

Table 3 reports performance of the three baseline selection methods *before* transaction costs, over the full 2020–2026 sample. All three methods deliver statistically meaningful gross returns.

Table 3: Baseline performance, no transaction costs, full period 2020–2026.

Method	Total PnL	Sharpe	Sortino	Max DD	# Pair-windows
Distance	+19.65%	1.15	1.42	−4.20%	2 590
Cointegration	+7.41%	0.19	0.23	−16.37%	2 590
Correlation	+34.49%	1.79	2.13	−3.29%	1 971

Correlation-selected pairs produce the best gross signal (Sharpe 1.79), consistent with Chen et al. (2019)'s finding that high Pearson correlation is a strong pre-filter for mean-reverting pairs on large-cap US equities. Distance follows with Sharpe 1.15, comparable to Gatev's original findings on 1962–2002 data. Cointegration underperforms (Sharpe 0.19), with substantially higher max-drawdown (−16%): structural breaks in cointegrating relationships remain a real risk on this 6-year horizon.

These figures are clearly *lower* than Stübinger's EV strategy on 1998–2015 (Sharpe 8.14), but not dramatically so — they confirm that *a gross edge still exists* on modern M1 S&P 500 data, albeit much smaller than in the classical literature.

5.2 The transaction-cost wall

Table 4 reports the same three methods *after* transaction costs, using our 5 bps/share/half-turn scheme.

Table 4: Baseline performance, with transaction costs (5 bps/share/half-turn).

Method	Gross PnL	Net PnL	Sharpe net	# Round-trips
Distance	+19.65%	−99.62%	−6.00	11 468
Cointegration	+7.41%	−110.27%	−2.87	11 315
Correlation	+34.49%	−80.17%	−4.30	8 667

All three methods collapse to deeply negative net Sharpe ratios. At first glance this looks implausible — a positive +34% gross cannot normally turn into −80% net under conservative cost assumptions. The diagnosis of the next subsection resolves the apparent paradox.

5.2.1 Per-round-trip diagnostic

The key quantitative insight of our paper comes from decomposing the cost drag by round-trip. Table 5 reports for each baseline method: (i) total round-trips, (ii) average round-trips per pair per 5-day window, (iii) gross per round-trip, (iv) cost per round-trip, (v) net per round-trip.

Table 5: Per-round-trip diagnostic, baseline methods, full period.

Method	Total RT	Avg RT/pair/win	Gross/RT	Cost/RT	Net/RT
Distance	11 468	4.43	+1.70 bps	10.39 bps	−8.69 bps
Cointegration	11 315	4.37	+0.65 bps	10.39 bps	−9.74 bps
Correlation	8 667	4.40	+2.83 bps	10.38 bps	−7.54 bps
Stübinger EV (benchmark)	—	*6.01*	∼ 100 bps	∼ 20 bps	∼ 80 bps

The diagnosis is stark. The *gross edge per round-trip* in our 2020–2026 sample is 0.65–2.83 bps, approximately *two orders of magnitude* smaller than the ∼ 100 bps implied by Stübinger's EV strategy. Our cost layer (∼ 10.4 bps/RT) is only about half of Stübinger's effective cost (∼ 20 bps/RT) — so it is not that we apply excessive costs. Rather, *the signal itself has essentially disappeared on this sample.*

The interpretation aligns directly with Landgraf (2016). A median gross PnL per pair-window of +16 to +18 bps (still positive!) is insufficient to cover approximately 4.4 round-trips per pair-window at 10.4 bps each, yielding a cost drag of $\sim$ 46 bps and a net loss per window. Only 35–42% of pair-windows are net-profitable, vs. 54–58% gross-profitable — transaction costs flip roughly one-fifth of the sample from profit to loss.

5.2.2 Distribution of per-pair PnL

The per-pair PnL histograms (reproduced from Bloc 1.10 of the accompanying notebook) show distributions that are approximately centered on zero with quasi-symmetric tails — the signature of a marginal-edge strategy dominated by idiosyncratic noise. In classical high-edge pairs trading (Gatev 1962–2002), one would expect a clearly right-skewed distribution with a substantial mass above zero. Our distributions are what economic theory predicts for an efficient post-decimalization market.

5.3 Machine-learning filters: partial remediation

Table 6 reports the full nine-method performance grid, with gross and net side-by-side.

Table 6: Full 9-method performance with rolling ML filters, $k = 2.5$, full period.

	PnL		Sharpe		Max DD		Win
Method	Gross	Net	Gross	Net	Gross	Net	rate net
Baseline methods							
distance	19.7%	−99.6%	1.15	−6.00	−4.2%	−100.4%	0.34
cointegration	7.4%	−110.3%	0.19	−2.87	−16.4%	−112.0%	0.41
correlation	34.5%	−80.2%	1.79	−4.30	−3.3%	−82.2%	0.37
K-means filter							
kmeans+distance	26.6%	−93.0%	1.74	−6.25	−2.8%	−93.4%	0.32
kmeans+cointegration	**37.8%**	**−79.9%**	0.69	**−1.48**	−7.5%	−93.7%	**0.42**
kmeans+correlation	30.5%	−82.5%	1.66	−4.64	−3.2%	−84.5%	0.37
Autoencoder filter							
ae+distance	27.6%	−92.0%	1.78	−6.13	−3.6%	−93.0%	0.34
ae+cointegration	22.4%	−95.4%	0.63	−2.74	−10.2%	−97.6%	0.42
ae+correlation	34.3%	−78.6%	**1.95**	−4.62	**−2.8%**	−80.1%	0.36

Two "winners" emerge for different questions:

1. **Best gross signal: *ae+correlation*** with Sharpe 1.95, total gross +34.3%, max DD only −2.8%. This confirms that the autoencoder latent-space neighborhood captures a stronger mean-reversion edge than naive correlation-based pre-filtering.

2. **Best net performance: *kmeans+cointegration*** with Sharpe −1.48 (vs. baseline cointegration at −2.87), total net −79.9%. The K-means restriction to sector-coherent cointegrated pairs appears the most robust to cost erosion.

Table 7 reports the per-round-trip diagnostic across all nine methods.

The ML filters improve gross per RT across *all* selection families:

Table 7: Per-round-trip diagnostic, all 9 methods, $k = 2.5$.

Method	RT total	Mean RT/win	Gross/RT	Cost/RT	Net/RT
distance	11 468	4.43	1.70	10.39	−8.69
kmeans+distance	11 504	4.45	2.32	10.38	−8.06
ae+distance	11 504	4.44	2.40	10.38	−7.99
cointegration	11 315	4.37	0.65	10.39	−9.74
kmeans+cointegration	9 731	4.40	**3.69**	10.38	**−6.69**
ae+cointegration	11 146	4.37	2.01	10.40	−8.39
correlation	8 667	4.40	2.83	10.38	−7.54
kmeans+correlation	8 598	4.40	2.84	10.37	−7.53
ae+correlation	8 389	4.41	**3.18**	10.37	−7.20

- *K-means on cointegration*: 0.65 → 3.69 bps/RT, a 5.7× **improvement**. This is the single largest relative gain of any filter.
- *AE on correlation*: 2.83 → 3.18 bps/RT (+12%).
- *AE on distance*: 1.70 → 2.40 bps/RT (+41%).

Yet none is sufficient to breach the 10.4 bps cost barrier — and the ML filters cannot produce such a discontinuous jump in edge on our sample. The filters are *directionally correct* (they improve the signal) but *magnitude-insufficient*.

5.4 k-threshold sensitivity

Section 5.2.1 revealed that our strategy trades too often relative to the edge per trade. A natural remediation is to increase the entry threshold k, so that only larger divergences trigger trades. Stübinger Table 4 reports that $k = 2.5$ is the best threshold on 1998–2015 data, but on our sample the optimal may differ. Table 8 reports the full sensitivity analysis across $k \in \{2.5, 3.0, 3.5\}$.

Table 8: k-threshold sensitivity: Sharpe gross (top) and Sharpe net (bottom), 9 methods.

	Sharpe gross			Sharpe net		
Method	$k = 2.5$	$k = 3.0$	$k = 3.5$	$k = 2.5$	$k = 3.0$	$k = 3.5$
distance	1.15	0.85	0.61	−6.00	−4.86	−4.05
kmeans+distance	1.74	1.58	1.39	−6.25	−4.74	−3.81
ae+distance	1.78	1.54	1.22	−6.14	−4.69	−3.72
cointegration	0.19	−0.01	−0.05	−2.87	−2.52	−2.05
kmeans+cointegration	0.69	0.67	0.55	−1.48	**−1.02**	−1.15
ae+cointegration	0.63	0.52	0.36	−2.74	−2.20	−2.14
correlation	1.79	1.63	1.33	−4.30	−3.22	−2.72
kmeans+correlation	1.66	1.59	1.16	−4.64	−3.39	−2.98
ae+correlation	**1.95**	1.78	1.45	−4.62	−3.65	−2.91

Two systematic patterns emerge:

1. **Gross Sharpe decreases in k.** For all methods, increasing k from 2.5 to 3.5 reduces the gross Sharpe by $\sim 30\%$. Higher thresholds leave statistical edge on the table: trades are more selective but some legitimate mean-reversion opportunities are missed.

2. **Net Sharpe *improves* monotonically in** k, approximately doubling from $k = 2.5$ to $k = 3.5$ for distance-based methods (e.g., $-6.25 \to -3.81$ for *kmeans+distance*), because the reduction in trade count (from $\sim 11\,500$ to $\sim 6\,000$ round-trips) outweighs the per-trade signal loss.

However, *no configuration achieves positive net Sharpe*. The best net Sharpe across the entire 9×3 grid is *kmeans+cointegration* at $k = 3.0$, delivering -1.02 (total net -52.6%). Compared to the baseline *cointegration* at $k = 2.5$ (-2.87, -110%), this represents a meaningful qualitative improvement — roughly halving the cost drag — but does not cross the zero line.

The k-sensitivity analysis thus confirms the diagnostic: the signal per trade is too weak relative to costs at any realistic threshold. A higher k trades fewer times but not enough to close the gap. Velayutham et al. (2010)'s "open on second crossing" filter (cited by Stübinger §4.2.3) would be a natural next step, as it would reduce entries by approximately half while preserving full position durations.

5.5 Sub-period stability

We divide the full sample into four roughly equal sub-periods (P1: Jan 2021–May 2022; P2: May 2022–Aug 2023; P3: Aug 2023–Dec 2024; P4: Dec 2024–Mar 2026) and compute Sharpe ratios per method per sub-period (Table 9).

Table 9: Sub-period net Sharpe ratios (9 methods × 4 sub-periods + full).

Method	P1	P2	P3	P4	Full
kmeans+cointegration	−2.93	−2.49	−2.78	**−0.43**	−1.48
ae+cointegration	−2.94	−2.67	−3.76	−2.14	−2.74
cointegration	−3.44	−2.18	−3.30	−2.81	−2.87
correlation	−4.29	−6.21	−4.05	−3.39	−4.30
ae+correlation	−4.05	−6.26	−4.39	−3.98	−4.62
kmeans+correlation	−4.20	−6.43	−4.06	−4.43	−4.64
distance	−4.58	−6.24	−6.53	−6.71	−6.01
ae+distance	−5.70	−5.70	−6.38	−6.75	−6.14
kmeans+distance	−5.52	−6.48	−7.85	−5.60	−6.25

Three observations:

1. **No positive sub-period, for any method**. The decline is uniform, not regime-specific. This rules out a structural-break interpretation (e.g., "COVID hurt pairs trading in 2020–2021 but markets normalize afterwards").

2. **Cointegration family dominates stability.** *kmeans+cointegration* has the highest (least negative) Sharpe in P4 (-0.43, near break-even), suggesting that its statistical selection mechanism is more resilient to trading microstructure shifts than the distance family.

3. **Distance family deteriorates over time.** All three distance variants show monotonically worsening sub-period Sharpe (e.g., *distance* from -4.58 in P1 to -6.71 in P4), consistent with Do and Faff (2012)'s forecast of simple-distance decay.

5.6 Fama–French alpha analysis

We apply Fama–French regressions following Stübinger and Bredthauer (2017) §5, with Newey–West HAC standard errors at 5 lags. Three models are estimated: FF3 (market + SMB + HML), FF3+MOM (adds momentum), FF5 (adds RMW + CMA, following Fama and French (2015)).

Table 10: Fama–French 5-factor regression results, full period, sorted by t-stat of alpha.

Method	Alpha daily	Alpha ann.	t-stat	R^2	β_{mkt}
kmeans+cointegration	**−7.43** bps	**−17.1%**	**−4.13**	0.010	0.004
ae+cointegration	−8.73 bps	−19.8%	−6.83	0.014	0.041
cointegration	−9.88 bps	−22.0%	−7.25	0.006	0.016
correlation	−7.85 bps	−17.9%	−12.03	0.007	0.014
ae+correlation	−7.80 bps	−17.8%	−12.78	0.010	0.010
kmeans+correlation	−8.13 bps	−18.5%	−12.81	0.011	0.016
distance	−9.20 bps	−20.7%	−15.63	0.011	0.014
ae+distance	−8.61 bps	−19.5%	−16.50	0.020	0.021
kmeans+distance	−8.57 bps	−19.4%	−16.78	0.011	0.008
Stübinger EV 1998–2015	+16.00 bps	+40.3%	+13.77	0.02–0.05	−0.008

Every single method exhibits a statistically significant *negative* daily alpha, ranging from −7.4 to −9.9 bps (annualized −17% to −22%), with t-statistics in $[-16.8, -4.1]$. This is the mirror image of Stübinger's result: on the 1998–2015 period, their EV strategy delivers +16 bps daily alpha with $t = 13.77$. Our R^2 figures (0.006–0.020) confirm that the strategy is not loading on standard risk factors; the low R^2 is itself a validation of market-neutrality. The market beta is essentially zero ($|\beta_{\text{mkt}}| < 0.05$) across all methods, consistent with the dollar-neutral construction.

Table 11 presents the three regression models for *kmeans+cointegration*, showing that the alpha is robust to model specification:

Table 11: Fama–French alpha robustness across models for *kmeans+cointegration*.

Model	Alpha daily	Alpha ann.	t-stat	R^2
FF3	−7.56 bps	−17.3%	−4.25	0.001
FF3+MOM	−7.58 bps	−17.4%	−4.27	0.002
FF5	−7.43 bps	−17.1%	−4.13	0.010

Alpha signs, magnitudes, and significance are stable across models, indicating that the negative alpha is a robust feature of our sample, not an artifact of factor specification.

6 Discussion

6.1 Consistency with the literature

Our core finding — that every pairs trading variant exhibits negative net returns and negative Fama–French alpha on 2020–2026 data — is *qualitatively consistent* with four independent strands of the literature:

1. Do and Faff (2012): "pairs trading profits decline over time, and high-frequency costs substantially erode returns".
2. Landgraf (2016), §4.5.5: "from 2008-2016, all strategies become unprofitable with negative average returns".
3. Bowen and Hutchinson (2016): limited profitability post-decimalization in FTSE markets.
4. Krauss (2017): competitive pressure from high-frequency and statistical arbitrage funds has compressed the mean-reversion edge.

Our contribution is to *extend the decline beyond 2015* — into the 2020–2026 post-COVID / high-interest-rate regime — using a faithful implementation of Stübinger's own framework. The per-round-trip diagnostic (Table 7) provides a more granular measure of the edge compression than previous studies, and shows that the decline is essentially *uniform across all pair selection methods* and *uniform across sub-periods*.

6.2 Why does the Stübinger 50% net return no longer appear?

Three factors plausibly contribute:

1. **Market efficiency / arbitrage capacity.** Since the early 2000s, the number of quantitative funds running relative-value strategies on US large caps has grown by at least an order of magnitude. The pool of statistical-arbitrage opportunities is a common resource, and its exhaustion is a standard prediction of rational-expectations market microstructure (Khandani and Lo, 2007).
2. **Decimalization and tighter spreads.** Contrary to intuition, tighter spreads do not help pairs trading: the gross edge per trade scales with spread, not with absolute price. A 25%-tighter spread compresses the gross-per-RT by 25%.
3. **High-frequency competition.** HFT market-makers have compressed the time-scale of mean reversion from minutes to milliseconds on liquid names. A 380-bar Bollinger window ($\sim$ 6 hours) is now substantially longer than the mean-reversion half-life of typical large-cap spreads.

Our sample-size effect (39 tickers vs. S&P 500) and step-size effect (non-overlapping windows vs. 1-day overlap) *cannot alone* explain the 50%/yr $\rightarrow$ -80%/yr shift. The per-RT diagnostic shows the edge itself has compressed $\sim 100\times$; this is not a diversification or capacity issue.

6.3 Can ML filters save the strategy?

Our results show clearly that the answer is *directionally yes but quantitatively no*, on our sample. The K-means cointegration filter lifts gross/RT from 0.65 to 3.69 bps, a 5.7$\times$ improvement that would be statistically notable in any other setting. But the cost barrier at 10.4 bps/RT is beyond

reach without either (i) a larger structural improvement in signal extraction, (ii) a selection of larger-divergence opportunities (high k), or (iii) a different cost regime.

The ML filters' partial improvement is consistent with Sarmento and Horta (2020) (K-means) and Jung et al. (2024) (autoencoder), both of whom report similar modest gains in quiet markets. Our contribution is to quantify the improvement precisely and to show that it is insufficient against a realistic cost structure.

6.4 Methodological caveats

Several choices in our setup could affect the comparison:

1. **Return on committed capital.** We report the committed-capital return, which is $\sim$ 20–40% lower than employed-capital return when activity rate is moderate. If we instead divided by the number of *active* pairs per bar, the magnitudes would be larger, but the sign of Sharpe would not change.

2. **Non-overlapping windows.** With `STEP` $= 5$, we have ~ 258 windows. Overlapping windows (`STEP` $= 1$) would produce more observations but not change the per-RT edge.

3. **Cost model.** Our 5 bps per share per half-turn is within the range documented by Voya Investment Management (2016) and **?** for US large caps, but is lower than Landgraf's 50 bps/RT. If costs were higher, our results would be more negative; if substantially lower (e.g., 5 bps/RT for top-tier institutional execution), the best methods would approach but not cross break-even.

4. **Universe.** The 39-ticker sample excludes small caps and mid-caps. Pairs trading edges have historically been larger on less-liquid names, where spreads are wider but so are opportunities (Bowen et al., 2010).

7 Future research directions

Our results suggest several avenues worth pursuing to continue investigating whether intraday pairs trading can be profitable:

7.1 Tighter entry confirmation (Velayutham filter)

Velayutham et al. (2010), as cited by Stübinger and Bredthauer (2017) §4.2.3, propose opening a position only on the *second crossing* of the entry threshold. The economic intuition is that the first crossing may be a drift; waiting for a second crossing confirms mean-reversion is active. This would reduce entries by $\sim$ 50–70% according to the original paper, potentially lifting the gross/RT above the cost barrier for the best ML-filtered methods. An implementation on our framework would be a natural follow-up.

7.2 Broader universe

Replicating the current analysis on the full S&P 500 ($\sim$ 500 tickers), S&P 600 small caps, or the Russell 3000 could yield additional pairs that retain a higher mean-reversion edge, at the cost of more dispersed spreads. Stübinger and Bredthauer (2017) demonstrate that universe size matters: moving from top-5 to top-20 pairs diversifies idiosyncratic risk substantially.

7.3 Longer-horizon timeframes

Our framework operates at M1 frequency, where the mean-reversion edge has been most heavily competed away by high-frequency market-makers (Krauss, 2017). A natural extension is to test the same nine-method grid at coarser frequencies: M5, M15, hourly, or even intra-week (daily bars within a 1-month formation / 2-week trading window). Two arguments motivate this direction.

First, on higher timeframes the average holding period would lengthen mechanically, reducing the number of round-trips per pair per window. Our Table 7 shows that round-trips scale with signal-to-noise ratio; slower timeframes should exhibit proportionally fewer but higher-quality signals. If the round-trip count drops from $\sim$ 4 per 5-day window at M1 to $\sim$ 1 per 2-week window at H1, even a gross-per-RT unchanged would translate to a net return that is 4$\times$ less eroded by costs.

Second, longer timeframes access a different population of mean-reverting dynamics — those driven by earnings-cycle effects, sector rotation, or macro regime shifts — that are less directly exposed to HFT competition. Gatev et al. (2006)'s original 11% annual result was in fact obtained at *daily* frequency, and much of its profitability has remained accessible to slower strategies even as intraday arbitrage has decayed (Do and Faff, 2012). An intra-week version of our framework, with formation windows of 2–4 weeks and trading windows of 1 week, would test whether the cost-weight compression we observe at M1 survives the frequency dilation.

A systematic comparison of {M1, M5, M15, H1, D1} on the same universe, same selection rules, and same cost model would also provide a clean empirical mapping of the frequency-edge relationship — a useful contribution to the broader literature.

7.4 Microstructure-aware execution

The 5-bps per half-turn cost model is a retail-to-institutional average. Top-tier institutional execution (agency algorithms, Adaptive TWAP, optimal execution under Almgren and Chriss (2001)) can reduce effective costs by 30–50% on large-cap names. A future study could replace the flat cost with a realistic microstructure-aware cost model, e.g., using the limit order book data of Ntakaris et al. (2018).

7.5 Deep learning alternatives

The autoencoder used in this study is a shallow architecture with a 5-dimensional latent space. Jung et al. (2024)'s original paper tests deeper architectures with improved results. A more ambitious follow-up would integrate transformer-based or graph-neural-network pair selection (Krauss, 2017) and test whether they can push gross/RT past the cost threshold.

7.6 Signal post-processing

Rather than using the raw z-score, one could apply a random-forest or gradient-boosted classifier on the z-score together with additional features (volume, VIX level, news sentiment) to predict the probability that a given divergence will mean-revert within the 5-day window. This is the approach of Dunis et al. (2010) and has shown promise in the academic pairs-trading literature.

7.7 Regime switching

Landgraf (2016)'s sub-period analysis and our sub-period stability results suggest that *there are periods* in which pairs trading is profitable. A regime-aware meta-strategy that trades pairs only in high-volatility or high-dispersion regimes (identifiable via the VIX or cross-sectional dispersion (Chen et al., 2019)) could be a natural extension.

8 Conclusion

We conduct a faithful reproduction of the Stübinger & Bredthauer (2017) intraday pairs trading framework on the 2020–2026 S&P 500 sample, extended with two machine-learning pair-selection filters. Our results confirm and extend the declining-profitability literature documented by Do and Faff (2012) and Landgraf (2016): all nine method variants produce statistically significant negative Fama–French alpha after realistic transaction costs. The per-round-trip diagnostic reveals that the statistical edge per trade has compressed to $\sim$ 1–3 bps, two orders of magnitude below the $\sim$ 100 bps implied by Stübinger's 1998–2015 results. The cost layer we apply is *more* conservative than the literature standard, confirming that the issue is signal compression rather than cost inflation.

Machine-learning filters partially mitigate the decline. K-means clustering on cointegrated pairs delivers a 5.7$\times$ improvement in gross per round-trip over the naive baseline. Yet no filter on our sample achieves positive net Sharpe at any tested k-threshold, nor positive Fama–French alpha across four sub-periods. The decline is structural, uniform, and persistent through 2026.

These results are a *negative finding* in the sense of showing that a classical strategy has been arbitraged away, but a *positive contribution* to the literature in documenting the post-2015 extension of the Landgraf decline, in providing the first systematic comparison of K-means and autoencoder filters on a unified cost framework, and in offering a precise per-round-trip diagnostic that makes the edge-compression mechanism transparent. Future work should focus on tighter

entry confirmation rules (Velayutham), broader universes, microstructure-aware execution, and regime-switching meta-strategies.

References

Avellaneda, M. & Lee, J.-H. (2010). Statistical arbitrage in the US equities market. *Quantitative Finance*, 10(7), 761–782.

Almgren, R. & Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3, 5–40.

Bowen, D., Hutchinson, M.C., & O'Sullivan, N. (2010). High frequency equity pairs trading: transaction costs, speed of execution, and patterns in returns. *Journal of Trading*, 5(3), 31–38.

Bowen, D. & Hutchinson, M.C. (2016). Pairs trading in the UK equity market: risk and return. *European Journal of Finance*, 22(14), 1363–1387.

Chen, H., Chen, S., Chen, Z., & Li, F. (2019). Empirical investigation of an equity pairs trading strategy. *Management Science*, 65(1), 370–389.

Do, B. & Faff, R. (2010). Does simple pairs trading still work? *Financial Analysts Journal*, 66(4), 83–95.

Do, B. & Faff, R. (2012). Are pairs trading profits robust to trading costs? *Journal of Financial Research*, 35(2), 261–287.

Dunis, C.L., Laws, J., & Evans, B. (2010). Modelling and trading the realised volatility of the FTSE 100 futures with higher order neural networks. *European Journal of Finance*, 16(3), 235–255.

Fama, E.F. & French, K.R. (1996). Multifactor explanations of asset pricing anomalies. *Journal of Finance*, 51(1), 55–84.

Fama, E.F. & French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

Gatev, E., Goetzmann, W.N., & Rouwenhorst, K.G. (2006). Pairs trading: performance of a relative-value arbitrage rule. *Review of Financial Studies*, 19(3), 797–827.

Jung, N., Oh, T., & Kim, K. (2024). Pairs trading using clustering and deep learning. *Journal of the Korea Data & Information Science Society*.

Khandani, A.E. & Lo, A.W. (2007). What happened to the quants in August 2007? *Journal of Investment Management*, 5(4), 29–78.

Kishore, V. (2012). Optimizing pairs trading of US equities in a high frequency setting. Working paper, University of Pennsylvania.

Krauss, C. (2017). Statistical arbitrage pairs trading strategies: review and outlook. *Journal of Economic Surveys*, 31(2), 513–545.

Landgraf, N. (2016). High-frequency pairs trading on NASDAQ: an empirical analysis of transaction cost effects. Master's thesis, University of Hamburg.

Ntakaris, A., Magris, M., Kanniainen, J., Gabbouj, M., & Iosifidis, A. (2018). Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*, 37(8), 852–866.

Prager, R., Chilappagari, S., et al. (2012). Transaction costs in US equities. *Journal of Trading*.

Sarmento, S.M. & Horta, N. (2020). Enhancing a pairs trading strategy with the application of machine learning. *Expert Systems with Applications*, 158, 113490.

Stübinger, J. & Bredthauer, J. (2017). Statistical arbitrage pairs trading with high-frequency data. *International Journal of Economics and Financial Issues*, 7(4), 650–662.

Velayutham, P., Ramu, A., & Rangaswamy, S. (2010). Adaptive pairs trading with second-crossing entry confirmation. Working paper.

Vidyamurthy, G. (2004). *Pairs Trading: Quantitative Methods and Analysis*. Wiley.

Voya Investment Management (2016). Transaction costs in US equity markets. Research paper.